Course Description	
COURSE NUMBER and NAME	BSIT 375 Knowledge Discovery and Data Science
UNITS	3
LENGTH OF CLASS	8 Weeks
COURSE DESCRIPTION	The course covers the process of automatically extracting valid, useful, and previously unknown information from data sources and using the information to make decisions. This course is designed to provide students with a solid understanding of the knowledge discovery process and the use of data mining concepts and tools as part of that process.
REQUIRED TEXT	Larose, Daniel T. (2014). *Discovering knowledge in data: An Introduction to data mining* (2nd ed.). Wiley-Blackwell. **Print ISBN:** 9780470908747, 0470908742 **eText ISBN:** 9781118975251, 1118975251
INSTRUCTIONAL METHOD	Online / On-Campus

Summary of Graded Work and Assessments

Graded work and assessments offer students the opportunity to show the degree of mastery for each CLO. The following table shows how assessments and CLOs align (link).

Assignments	Totals	Weight	CLOs
Engagement and Professionalism - Laboratory Class Activities(Rubric)	160	16%	1, 2, 3, 4, 5
Week 1 Discussion	50	5%	1, 2
Week 2 Discussion	50	5%	1, 2, 3
Week 3 Discussion	50	5%	1, 2, 3, 4
Week 3 Assignment	90	9%	1, 2, 3, 4
Week 4 Discussion	50	5%	1, 2, 3, 4, 5
Week 5 Discussion	50	5%	1, 2, 3
Week 5 Assignment	100	10%	1, 2, 3
Week 6 Discussion	50	5%	1, 2, 3, 4, 5, 6
Week 7 Discussion	50	5%	1
Week 7 Assignment	100	10%	1, 2, 3, 4
Week 8 Project Demo and Presentation	200	20%	1, 2, 3, 4, 5, 6
Total Points/Percentage	**1000 Points**	**100%**	

Course Policies

For Westcliff's course policies, please see the Course Policies document.

Discussion Requirements

For all discussions, the primary response is due by Wednesday at 11:59 p.m. Pacific Time. The primary response must be at least 200 words in length and fully address the topic, demonstrating critical thinking and understanding. Each student must then also post a minimum of two responses to other students in the discussion by Saturday night at 11:59 p.m. Pacific Time. Each peer response must be at least 50 words in length and substantively engage with the other student's original post, continuing the discussion in a professional manner. If at any time information or material is brought in from an outside source or website, it must be properly cited following APA 7th edition guidelines, and a full reference must be provided.

Assignment Requirements

Each assignment deliverable is specifically defined in the assignment instructions, such as page length, citations and references, audio or video, presentations, tables, etc. For all written assignments, the required page length does not include the cover or references pages. Refer to the specific requirements as stated in each assignment, and reach out to your instructor for additional information as needed. All graded submissions are due by Sunday at 11:59 p.m. Pacific Time.

All written work must adhere to APA 7th edition academic formatting requirements including core components such as the cover page, page numbers, headings, citations, 1" margins, paragraph indentations, left alignment, double spacing throughout, and the final references using hanging indents.

Participation Requirements

Students are required to attend each live class session either in person or virtually as stipulated in the course policies. Participation in the live class session is determined by actively engaging, answering or asking questions, providing comments, interacting in group activities, etc., as required by the instructor. Students who are unable to attend the live in-class or virtual sessions must follow the Virtual Class Session (VCS) submission requirements as stated in the Course Policies document.

Writing Center

The Westcliff University Writing Center is dedicated to providing quality support to students and faculty. From assignment review, to in-class workshops, to dissertation support, to publication help, the Writing Center is committed to empowering individuals to use the written language to articulate and disseminate knowledge.

Course Learning Outcomes (CLOs)

Learning outcomes are statements that describe significant and essential scholarship that students have achieved and can reliably demonstrate at the end of the course. Learning outcomes identify what the learner will know and be able to do by the end of a course – the essential and enduring knowledge, abilities (skills), and attitudes (values, dispositions) that constitute the integrated learning needed for successful completion of this course. The learning outcomes for this course summarize what students can expect to learn, and how this course is tied directly to the educational outcomes of the degree.

Course Learning Outcomes (CLOs)	PLOs
1. Use appropriate tools to prepare datasets, impute missing data and program them.	2, 6, 7
2. Define all the knowledge discovery models and concepts and recognize them in examples and case studies.	2, 6, 7
3. Summarize the assumptions and conditions for model set up and implementation for linear and logistic regression, K-means clustering, networks, decision trees, association.	4, 6, 7
4. Demonstrate how to perform all statistical calculations and set up all the models in real datasets for knowledge discovery.	4, 5
5. Implement regularization to prevent models from overfitting the training data.	4, 5, 6
6. Develop model evaluation strategies.	5

Detailed Course Outline

The following outline provides important assignment details for this course, unit by unit. Students are responsible for all of the assignments given. Please refer to the Detailed Description of Each Grading Criteria in the syllabus for specific information about each assignment.

Week 1

Assignments to complete this week:

- Reading: Discovering Knowledge in Data: An Introduction to Data Mining (2nd ed.)
  - Chapter 1: An Introduction to Data Mining
  - Chapter 2: Data Pre-Processing
  - Chapter 3: Exploratory Data Analysis

Week 1 In-Class Activity (Rubric)

This week's live class activity comprises an individual task for each student, focusing on setting up the environment for Data Mining. Students will be guided through the following tasks:

- Environment setup: The instructor will guide students to choose programming languages like R or Python. The rest of the course will follow the same programming language for exercises.
- Read and practice the exercises listed in the section "R Zone" of Chapters 2 and 3.

The instructor will assist in selecting tools, programming language, libraries, and examples to be utilized. After completion, students are expected to share their findings with the live class.

Week 1 Discussion (Rubric)

1. CRISP-DM is not the only standard process for data mining. Research an alternative methodology (Hint: SEMMA, from the SAS Institute). Discuss the similarities and differences with CRISP-DM. Use all the resources available for search at Westcliff University and on the Internet.
2. Use the *churn* data set on the book series website for the following exercises.
   a. Explore whether there are missing values for any of the variables.
   b. Compare the area code and state fields. Discuss any apparent abnormalities.
   c. Use a graph to visually determine whether there are any outliers in the number of calls to customer service.
3. Use the *Adult* data set from the book series website for the following exercises. The target variable is *income,* and the goal is to classify income based on the other variables.
   a. Which variables are categorical and which are continuous?
   b. Using software, construct a table of the first 10 records of the data set, in order to get a feel for the data.
   c. Investigate whether we have any correlated variables.
   d. For each of the categorical variables, construct a bar chart of the variable, with an overlay of the target variable. Normalize if necessary.
   e. Discuss the relationship, if any, each of these variables has with the target variables.

f. Which variables would you expect to make a significant appearance in any data mining classification model we work with?

Post your analysis in the discussion board and then review and comment on another submission.

This is a learning and application exercise designed to give you an opportunity to apply concepts learned in a pragmatic and meaningful way that will enable you to gain valuable and relevant knowledge in an effort to augment your skill set and enhance your professional career.

Team Project – Team Selection

Instead of an exam in week 6, there will be a team project. This information is available in Week 8. For week one, teams of 3-4 students will be selected voluntarily or as assigned by your professor. Each team member will be responsible for the overall project, just as a project team in a business situation would be.

Although nothing beyond identifying the teams is due for week one, teams are encouraged to begin reviewing the project information and determining what role each person will take as an IT analyst and who will be responsible for each of the required portions. Please note that the distribution of the work must be even, both in the work to complete the project as well as during the presentation.

Week 2

Assignments to complete this week:

- Reading: Discovering Knowledge in Data: An Introduction to Data Mining (2nd ed.)
  - Chapter 4: Univariate Statistical Analysis
  - Chapter 5: Multivariate Statistical Analysis
  - Chapter 6: Preparing to Model the Data

Week 2 Live Class Activity (Rubric)

This week's live class activity involves group work. Students will be guided through the following tasks:

- Read and practice the exercises listed in the section "R Zone" of Chapters 4, 5, and 6.

The instructor will assist in dividing the class into different groups based on the class size, selecting tools, libraries, and a sample dataset to be utilized. After completion, students are required to share their findings with the live class.

Week 2 Discussion (Rubric)

1. Explain what is meant by statistical inference. Give an example of statistical inference from everyday life.
2. Using the theory in chapter 4, consider: in a sample of 100 customers, 240 churned when the company raised rates. Test whether the population proportion of churners is less than 25%, using a level of significance $\alpha = 0.01$.
3. Use the *cereals* data set included in the book series website for the following exercises. Use regression to estimate the *rating* based on *fiber* alone. Then, answer:
   a. What is the estimated regression equation?
   b. Explain clearly the value of the slope coefficient you obtained in the regression.
   c. What does the value of the y-intercept mean for the regression equation you obtained? Does it make sense in this example?
   d. What would be a typical prediction error obtained from using this model to predict *ratings?* Which statistic are you using to measure this? What could we do to lower this estimated prediction error?
   e. How closely does our model fit the data? Which statistic are you using to measure this?
   f. Find a point estimate for the rating for a cereal with a fiber content of 3 grams.
   g. Find a 95% confidence interval for the true mean rating for all cereals with a fiber content of 3 grams.
   h. Find a 95% prediction interval for a randomly chosen cereal with a fiber content of 3 grams.
   i. Based on the regression results, what would we expect a scatter plot of *rating* versus *fiber* to look like? Why?

Then, consider a multiple regression to estimate the *rating* based on *fiber* and *sugars.*

   a. What is the estimated regression equation?

b. Explain clearly and completely the value of the coefficient for fiber you obtained in the regression.
c. Compare the r^2 values from the multiple regression and the regression done earlier in the exercises. What is going on? Will this always happen?
d. Compare the s values from the multiple regression and the regression done earlier in the exercises. Which value is preferable, and why?

4. Answer the following:
   a. Explain the difference between supervised and unsupervised methods. Which data mining tasks are associated with unsupervised methods? Supervised? Both?
   b. Describe the differences between the training set, test set, and validation set. Should we strive for the highest possible accuracy with the training set? Why or why not? How about the validation set?
   c. How is the bias–variance trade-off related to the issue of overfitting and underfitting? Is high bias associated with overfitting and underfitting, and why? High variance?
   d. Explain why we sometimes need to balance the data.
   e. Suppose we are running a fraud classification model, with a training set of 10,000 records of which only 400 are fraudulent. How many fraudulent records need to be resampled if we would like the proportion of fraudulent records in the balanced data set to be 20%?

Post your analysis in the discussion board and then review and comment on another submission.

Team Project:

All students are to form groups of 3-4 by the end of Week 2. Each group will work on the team project throughout the entire session and submit the results in Week 8 for grading.
Before the end of Week 2, the team must meet (virtually or in person) to discuss the Week 8 team project (PA) and create the Project Team Charter. The completed charter identifies the team members, each member's responsibilities, and the deadline for each member's analysis. The team can submit a draft of the analysis and presentation in Week 6, for faculty non-graded feedback during Week 7.

Project Team forms and information can be found here.

Team Project Description:

Using everything you learned in this class, analyze the *churn* data and answer the following exercises. Make sure that the correlated variables have been accounted for.

1. Apply a CART model for predicting *churn.* Use default misclassification costs. Determine the following measures.
   a. Proportion of false positives.
   b. Proportion of false negatives.
   c. Overall error rate.
   d. Overall model accuracy (1 – overall error rate).

   e. Sensitivity
   f. Specificity
2. In a typical churn model, in which interceding with a potential churner is relatively cheap but losing a customer is expensive, which error is more costly: a false negative or a false positive (where positive = customer predicted to churn)? Explain.
3. Based on your answer to Exercise 2, adjust the misclassification costs for your CART model to reduce the prevalence of the more costly type of error. Rerun the CART algorithm. Compare the false positive, false negative, sensitivity, specificity, and overall error rate with the previous model. Discuss the trade-off between the various rates in terms of cost for the company.
4. Perform a cost/benefit analysis for the default CART model from exercise 1 as follows. Assign a cost or benefit in dollar terms for each combination of false and true positives and negatives, similar to **Table 14.4**. Then, using the contingency table, find the overall anticipated cost.
5. Perform a cost/benefit analysis for the CART model with the adjusted misclassification costs. Use the same cost/benefits assignments as for the default model. Find the overall anticipated cost. Compare with the default model, and formulate a recommendation as to which model is preferable.
6. Construct a lift chart for the default CART model. What is the estimated lift at 20%? 33%? 40%? 50%?
7. Construct a gains chart for the default CART model. Explain the relationship between this chart and the lift chart.
8. Construct a lift chart for the CART model with the adjusted misclassification costs. What is the estimated lift at 20%? 33%? 40%? 50%?
9. Construct a single lift chart for both of the CART models. Which model is preferable over which regions?
10. Now turn to a C4.5 decision tree model, and redo Exercises 1–9. Compare the results. Which model is preferable?
11. Next, apply a neural network model to predict churn. Calculate the following measures.
    a. Proportion of false positives.
    b. Proportion of false negatives.
    c. Overall error rate.
    d. Overall model accuracy (1 – overall error rate).
    e. Sensitivity
    f. Specificity
12. Construct a lift chart for the neural network model. What is the estimated lift at 20%? 33%? 40%? 50%?
13. Construct a single lift chart that includes the better of the two CART models, the better of the two C4.5 models, and the neural network model. Which model is preferable over which regions?
14. In view of the results obtained above, discuss the overall quality and adequacy of our *churn* classification models.

Each team member needs to take a designated role, and will contribute work and present based upon that role. One person should be the Project Manager, while another can focus on the physical network requirements, and another on the software requirements. All will require including different statistical analyses in your portion of the work and presentation.

Week 3

Assignments to complete this week:

- Reading: Discovering Knowledge in Data: An Introduction to Data Mining (2nd ed.)
  - Chapter 7: k-Nearest Neighbor Algorithm
  - Chapter 10: Hierarchical and K-Means Clustering

Week 3 Live Class Activity (Rubric)

This week's live class activity involves group work. Students will be guided through the following tasks:

- Read and practice the exercises listed in the section "R Zone" of Chapters 7 and 10.

The instructor will assist in dividing the class into different groups based on the class size, selecting tools, libraries, and a sample dataset to be utilized. After completion, students are required to share their findings with the live class.

Week 3 Discussion (Rubric)

1. Using the *ClassifyRisk* data set with predictors *age, marital status*, and *income*, and target variable *risk*, find the *k*-nearest neighbor for record #1, using $k = 2$ and Euclidean distance.
2. Use the *cereals* data set, included at the book series website, for the following questions. Make sure that the data are normalized.
   a. Using all of the variables except *name* and *rating,* run the *k*-means algorithm with $k = 5$ to identify clusters within the data.
   b. Develop clustering profiles that clearly describe the characteristics of the cereals within the cluster.
   c. Rerun the *k*-means algorithm with $k = 3$.
   d. Which clustering solution do you prefer, and why?
   e. Develop clustering profiles that clearly describe the characteristics of the cereals within the cluster.
   f. Use cluster membership to predict *ratings.* One way to do this would be to construct a histogram of *ratings* based on cluster membership alone. Describe how the relationship you uncovered makes sense, based on your earlier profiles.

Post your analysis in the discussion board and then review and comment on another submission.

Week 3 Assignment (Rubric)

Write a 3-4 page taxonomy of various types of machine learning algorithms, formatted in APA style, including relevant figures. Additionally, provide at least two real-world case scenarios where these algorithms are applied, highlighting their strengths and weaknesses.

Week 4

Assignments to complete this week:

- Reading: Discovering Knowledge in Data: An Introduction to Data Mining (2nd ed.)
  - Chapter 8: Decision Trees
  - Chapter 12: Association Rules

Week 4 Live Class Activity (Rubric)

This week's live class activity involves group work. Students will be guided through the following tasks:

- Read and practice the exercises listed in the section "R Zone" of Chapters 8 and 12.

The instructor will assist in dividing the class into different groups based on the class size, selecting tools, libraries, and a sample dataset to be utilized. After completion, students are required to share their findings with the live class.

Week 4 Discussion (Rubric)

1. Use the *churn* data set available at the book series website and answer the following questions. Normalize the numerical data and deal with the correlated variables.
   a. Generate a CART decision tree.
   b. Generate a C4.5-type decision tree.
   c. Compare the two decision trees and discuss the benefits and drawbacks of each.
   d. Generate the full set of decision rules for the CART decision tree.
   e. Generate the full set of decision rules for the C4.5 decision tree.
   f. Compare the two sets of decision rules and discuss the benefits and drawbacks of each.
2. Use the *churn* data set for the following exercises. Use the Churn_Training_File. Filter out all variables except the following: VMail Plan, Intl Plan, CustServ Calls, and Churn. Set CustServ Calls to be ordinal. Allow the three predictors to be in either antecedent or consequent, but do not allow Churn to be in the antecedent.
   a. Set the minimum antecedent support to 1%, the minimum rule confidence to 5%, and the maximum number of antecedents to 1. Use rule confidence as your evaluation measure.
   b. Find the association rule with the greatest lift.
   c. Report the following for the rule in (a).
      i. Number of instances, Support % (as defined in this chapter), Confidence %, Rule support %, Lift, Deployability
   d. Explain, in terms of this data, what each of the measures in (c) means.
   e. Set the minimum antecedent support to 1%, the minimum rule confidence to 5%, and the maximum number of antecedents to 1.
   f. Generate rules using confidence difference as your evaluation measure, with evaluation measure lower bound = 40. Explain what this evaluation measure means.

g. For the rules that are generated, use hand calculations to compute the reported evaluation measure, and show that the evaluation measure lower bound has been met.

Post your analysis in the discussion board and then review and comment on another submission.

Week 5

Assignments to complete this week:

- Reading: Discovering Knowledge in Data: An Introduction to Data Mining (2nd ed.)
  - Chapter 9: Neural Networks
  - Chapter 11: Kohonen Networks

Week 5 Live Class Activity (Rubric)

This week's live class activity involves group work. Students will be guided through the following tasks:

- Read and practice the exercises listed in the section "R Zone" of Chapters 9 and 11.

The instructor will assist in dividing the class into different groups based on the class size, selecting tools, libraries, and a sample dataset to be utilized. After completion, students are required to share their findings with the live class.

Week 5 Discussion (Rubric)

Welcome to week 5 of your course. This discussion question will help you **prepare for your Week 7 assignment,** *and as such, will have a fair amount of detail. Read the week 7 assignment listed in week 7 of the course*. Then, please **provide an outline that itemizes the concepts that you will include in your Week 7 submissions**. Please be sure to *include concepts learned in the course and information (findings, conclusions) from your **earlier assignments** (if applicable) papers.* Provide some brief details for each item that is outlined. Please keep in mind that you should have place-holders for material not yet covered in lectures from weeks 6 and 7. Here is an itemized list that summarizes the requirements of this DQ (*include every item in the bullet list below, or you will not receive full credit)*:

- Outline that itemizes concepts learned in the course
- Include information (findings, conclusions) from your earlier assignments (if applicable)
- Placeholders for material not yet covered (taken from syllabus)
- Brief details for each item that is outlined (creating a comprehensive outline)

Week 5 Assignment (Rubric)

1. Use the data set *churn.* Normalize the numerical data, recode the categorical variables, and deal with the correlated variables. Answer the following:
   a. Generate a neural network model for classifying *churn* based on the other variables. Describe the topology of the model.
   b. Which variables, in order of importance, are identified as most important for classifying *churn?*
   c. Compare the neural network model with the CART and C4.5 models for this task in **Chapter 6**. Describe the benefits and drawbacks of the neural network model

compared to the others. Is there convergence or divergence of results among the models?

2. Use the *ClassifyRisk* data set.
    a. Run an NN model predicting income based only on age. Use the default settings and make sure there is one hidden layer with one neuron.
    b. Consider the following quantity: (weight for Age-to-Neuron1) + (weight for Bias-to-Neuron1) * (weight for Neuron 1-to-Output node). Explain whether this makes sense, given the data, and why.
    c. Make sure the target variable takes the flag type. Compare the sign of (weight for Age-to-Neuron 1) + (weight for Bias-to-Neuron 1) * (weight for Neuron 1-to-Output node) for the good risk output node, as compared to the bad loss output node. Explain whether this makes sense, given the data, and why.
3. Use the *churn* data set for the following exercises. Use the Churn_Training_File. Filter out all variables except the following: VMail Plan, Intl Plan, CustServ Calls, and Churn. Set CustServ Calls to be ordinal. Allow the three predictors to be in either antecedent or consequent, but do not allow Churn to be in the antecedent.
    a. Set the minimum antecedent support to 1%, the minimum rule confidence to 5%, and the maximum number of antecedents to 1. Use rule confidence as your evaluation measure.
    b. Find the association rule with the greatest lift.

    Report the following for the rule in (a): Number of instances, Support % (as defined in this chapter), Confidence %, Rule support %, Lift, Deployability

    c. Explain, in terms of this data, what each of the measures in (c) means.
    d. Set the minimum antecedent support to 1%, the minimum rule confidence to 5%, and the maximum number of antecedents to 1.
    e. Generate rules using confidence difference as your evaluation measure, with evaluation measure lower bound = 40. Explain what this evaluation measure means.
    f. For the rules that are generated, use hand calculations to compute the reported evaluation measure, and show that the evaluation measure lower bound has been met.
    g. Generate rules using confidence difference as your evaluation measure, with evaluation measure lower bound = 30.
    h. Select the rule with the highest deployability. Explain why the deployability of this rule is greater than the rule we found in Question 1a.
    i. Set the minimum antecedent support to 1%, the minimum rule confidence to 5%, and the maximum number of antecedents to 1.
    j. Generate rules using confidence ratio as your evaluation measure, with the evaluation measure lower bound = 40. Explain what this evaluation measure means.
    k. Select the rule involving Intl Plan. Use hand calculations to compute the reported evaluation measure, and show that the evaluation measure lower bound has been met.

l. Compare the results from Exercise 13 with the results from the EDA and decision tree analysis in **Chapters 3** and **6**. Discuss similarities and differences. Which analysis format do you prefer? Do you find a confluence of results?
m. Apply the GRI algorithm to uncover association rules for predicting either churn or non churn behavior. Specify reasonable lower bounds for support and confidence.
n. Compare the results from the a priori algorithm with those of the GRI algorithm. Which algorithm yields a richer set of rules, and why? Which algorithm is probably preferable for this particular data set? Why?

Week 6

Assignments to complete this week:

- Reading: Discovering Knowledge in Data: An Introduction to Data Mining (2nd ed.)
    - Chapter 13: Imputation of Missing Data

Week 6 Live Class Activity (Rubric)

This week's live class activity involves group work. Students will be guided through the following tasks:

- Read and practice the exercises listed in the section "R Zone" of Chapter 13.

The instructor will assist in dividing the class into different groups based on the class size, selecting tools, libraries, and a sample dataset to be utilized. After completion, students are required to share their findings with the live class.

Week 6 Discussion (Rubric)

Before proceeding, use the *cereals* data set for Exercises 7–12. Report the standard error of each imputation.

1. Impute the potassium content of Almond Delight using multiple regression.
2. Impute the potassium content of Cream of Wheat.
3. Impute the carbohydrate value of Quaker Oatmeal.
4. Impute the sugar value of Quaker Oatmeal.
5. Insert the value obtained in Exercise 10 for the sugar value of Quaker Oatmeal, and impute the carbohydrate value of Quaker Oatmeal.
6. Compare the standard errors for the imputations obtained in Exercises 9 and 11. Explain what you find.

Post your analysis in the discussion board and then review and comment on another submission.

Project – Draft submission

Please review the Project Information in Week 8.

Prepare a draft for submission by Sunday evening at 11:59 pm of week 6. The faculty member will provide feedback on the draft for content only during week 7. You will then use that feedback to improve and finalize the project for presentation and submission during the last class meeting in week 8.

Week 7

Assignments to complete this week:

- Reading: Discovering Knowledge in Data: An Introduction to Data Mining (2nd ed.)
    - Chapter 14: Model Evaluation Techniques

Week 7 Live Class Activity (Rubric)

This week's live class activity involves group work. Students will be guided through the following tasks:

- Read and practice the exercises listed in the section "R Zone" of Chapter 14.

The instructor will assist in dividing the class into different groups based on the class size, selecting tools, libraries, and a sample dataset to be utilized. After completion, students are required to share their findings with the live class.

Week 7 Discussion (Rubric)

Read this article Transforming Ordinal Predictors to Numeric for Binary Logistic Models, identify the theory concepts you learned in this topic, and how these were used. For the three (3) concepts you identified, write a paragraph each summarizing the:

1. Concept definition
2. The problem with this concept was used in.
3. Learnings you got from using and calculating the concept.
4. Identify challenges you encountered and how you solved them.

Post your analysis in the discussion board and then review and comment on another submission.

Week 7 Assignment (Rubric)

In this assignment, students need to answer the following questions and compile them, and submit. Remember, you need to provide relevant examples and show the actual scenario.

1. Describe what we should do if there are many variables with many missing values.
2. Think of a data set where a potential pattern in missingness would represent good information.
3. Using all resources at Westcliff University and the Internet, search and summarize the methods for handling patterns in missingness.

Week 8

Assignments to complete this week:

- Reading
  - Review all the chapters from weeks 1 to 7.

Week 8 Team Project Presentation (Rubric)

Welcome to the last week of this course. Every group of 3-4 students presents its analysis of the Team Project. Presentations are to be summarized in a professional PowerPoint with a minimum of 18 slides containing the following sections:

1. Title
2. Executive Summary
3. Problem/challenge/opportunity studied with a list of clear analysis questions
4. For each question identified at point 3, prepare one slide that shows how the question was solved using all the statistical concepts learned throughout the class. Clearly identify the statistic used, how it was implemented/calculated in SAS.
5. Summary of all findings and impact on the business.
6. Proposed further analysis
7. References

Note that presentations should not exceed 20 minutes.

Team Project Description:

Using everything you learned in this class, analyze the *churn* data and answer the following exercises. Make sure that the correlated variables have been accounted for.

1. Apply a CART model for predicting *churn.* Use default misclassification costs. Determine the following measures.
   a. Proportion of false positives.
   b. Proportion of false negatives.
   c. Overall error rate.
   d. Overall model accuracy (1 – overall error rate).
   e. Sensitivity
   f. Specificity
2. In a typical churn model, in which interceding with a potential churner is relatively cheap but losing a customer is expensive, which error is more costly: a false negative or a false positive (where positive = customer predicted to churn)? Explain.
3. Based on your answer to Exercise 2, adjust the misclassification costs for your CART model to reduce the prevalence of the more costly type of error. Rerun the CART algorithm. Compare the false positive, false negative, sensitivity, specificity, and overall error rate with the previous model. Discuss the trade-off between the various rates in terms of cost for the company.
4. Perform a cost/benefit analysis for the default CART model from exercise 1 as follows. Assign a cost or benefit in dollar terms for each combination of false and true positives

and negatives, similar to **Table 14.4**. Then, using the contingency table, find the overall anticipated cost.

5. Perform a cost/benefit analysis for the CART model with the adjusted misclassification costs. Use the same cost/benefits assignments as for the default model. Find the overall anticipated cost. Compare with the default model, and formulate a recommendation as to which model is preferable.
6. Construct a lift chart for the default CART model. What is the estimated lift at 20%? 33%? 40%? 50%?
7. Construct a gains chart for the default CART model. Explain the relationship between this chart and the lift chart.
8. Construct a lift chart for the CART model with the adjusted misclassification costs. What is the estimated lift at 20%? 33%? 40%? 50%?
9. Construct a single lift chart for both of the CART models. Which model is preferable over which regions?
10. Now turn to a C4.5 decision tree model, and redo Exercises 1–9. Compare the results. Which model is preferable?
11. Next, apply a neural network model to predict churn. Calculate the following measures.
    a. Proportion of false positives.
    b. Proportion of false negatives.
    c. Overall error rate.
    d. Overall model accuracy (1 – overall error rate).
    e. Sensitivity
    f. Specificity
12. Construct a lift chart for the neural network model. What is the estimated lift at 20%? 33%? 40%? 50%?
13. Construct a single lift chart that includes the better of the two CART models, the better of the two C4.5 models, and the neural network model. Which model is preferable over which regions?
14. In view of the results obtained above, discuss the overall quality and adequacy of our *churn* classification models.

Each team member needs to take a designated role, and will contribute work and present based on that role. One person should be the Project Manager, while another can focus on the physical network requirements, and another on the software requirements. All will require including different statistical analyses in your portion of the work and presentation.